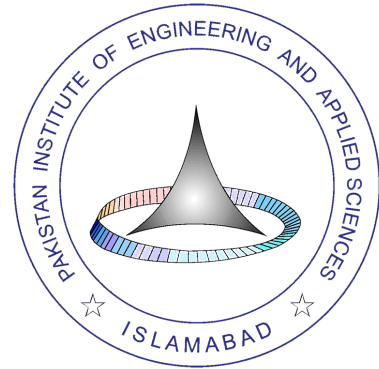


Basics of Geant-4 and Corsika

Simulation technique to study ionization of Cosmic rays



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Abstract

During this internship, I was introduced to two important simulation tools used in modern physics research: Geant4 and CORSIKA 7.8010. My work with Geant4 focused on understanding its basic structure, learning how to set up and run simulations, and studying the example B1 project, which provided practical experience in defining detector geometry, running particle interactions, and visualizing results. In the second part of the internship, I worked with CORSIKA on a Linux environment. I learned the fundamentals of Linux commands, the installation process of CORSIKA, and how to run basic cosmic ray air shower simulations. This provided me with a foundation in handling large-scale physics simulation tools and improved my skills in both particle physics concepts and computational techniques.

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Part I

Geant-4

Chapter 1

Introduction to Geant-4

1.1 | Introduction to Monte Carlo Simulations

Monte Carlo simulations are computational methods that rely on random sampling to approximate the behavior of complex systems. Instead of solving equations directly, the method generates a large number of random events according to the probability laws of the process being studied. By repeating this process many times, the simulation builds up a statistical picture of the system. This approach is especially useful in physics, where interactions often involve a wide range of possible outcomes that are too complicated to calculate analytically.

In particle and nuclear physics, Monte Carlo simulations are essential for modeling phenomena such as particle scattering, energy loss in matter, and detector responses. By simulating millions of particle histories, scientists can predict experimental outcomes, design detectors, and compare theoretical models with observed data. These simulations reduce the need for costly experiments and allow researchers to explore scenarios that may be impractical or impossible to reproduce in the laboratory.

1.2 | Introduction to Geant4

Geant4 (GEometry ANd Tracking) is an open-source simulation toolkit designed to model the passage of particles through matter. It is one of the most widely used Monte Carlo frameworks in physics research and engineering, especially in high-energy physics, nuclear science, medical physics, and space research. The toolkit allows researchers to construct complex detector geometries, define materials, and simulate particle beams interacting with those materials. Because it is modular, users can customize physics processes, visualization, and data collection according to their needs, whether for large-scale experiments at facilities like CERN or smaller academic projects.

A major advantage of Geant4 is its broad coverage of physics processes, including electromagnetic, hadronic, optical, and radioactive decay interactions. The software provides powerful visualization tools for displaying detector geometries and particle trajectories in real time, and it can output detailed information such as energy deposition and secondary particle production. Written in C++ and based on object-oriented programming, Geant4 is both flexible and extensible, making it suitable for simulations ranging from fundamental research to practical applications like radiation shielding and medical treatment planning.

1.3 | Structure of a Geant4 Project

A typical Geant4 project follows a standard folder and file organization that makes the simulation easy to compile, run, and maintain. At the root of the project lies the CMakeLists.txt file, which is essential for configuring and building the project using CMake. The src folder contains all the source code files (.cc) that define classes such as the detector construction, physics list, and primary particle generator. The include folder holds the corresponding header files (.hh), which declare the classes and functions used throughout the simulation.

The build folder is usually kept empty at the start and is the location where all compilation outputs are generated. This separation ensures that the source code remains clean while CMake generates project-specific build files. Optional macro files, such as vis.mac, are also an important part of a Geant4 project. These files contain pre-written commands that control visualization and simulation parameters, making it easier to rerun simulations with consistent settings.

The general workflow of a Geant4 project can be summarized as: define, build, run, and visualize. First, the user defines the detector geometry, materials, and physics processes. Next, the project is configured and built using CMake and a compiler (e.g., MSVC on Windows). Once built, the compiled executable is run, often with macro files to automate commands. Finally, visualization tools are used to display particle trajectories and detector volumes, providing both qualitative insights and quantitative data for analysis.

The folder structure of a typical Geant4 project is shown below:

YourProject/	
CMakeLists.txt	Main CMake configuration file
src/	All your .cc files
include/	All your .hh files
build/	Empty folder where build files go
vis.mac	Optional macro file for visualization

Chapter 2

Example B1 Study

2.1 | Setting Up Geant4 for Example B1

Setting up Geant4 to run the B1 example requires preparing both the software environment and the build system. The first step is to install the necessary tools, which include the Visual Studio Build Tools (with MSVC compiler support) and CMake for managing the build process. These tools ensure that the C++ code of Geant4 and its examples can be compiled correctly on Windows. In addition, the Windows 10/11 SDK must be available to provide the system libraries required for compilation.

Once the prerequisites are installed, the Geant4 environment needs to be configured. This is typically done by running the environment setup script (`geant4.bat`), which defines paths to the Geant4 libraries, data files, and executables. To carry out compilation, the x64 Native Tools Command Prompt for VS 2022 is used, as it provides the correct compiler environment for 64-bit builds. Inside this prompt, the CMake command is executed to configure the project in an empty build/ folder, linking it with the source code of the B1 example. After configuration, the `cmake -build .` command is run to compile the project and generate the executable for the B1 simulation.

2.2 | What is B1 Example?

The B1 example is one of the simplest tutorials provided with Geant4 and is designed for new users. Its geometry consists of a very basic detector setup, usually represented by a box-shaped volume. This simplicity allows beginners to focus on understanding how a Geant4 project is structured and how different components—such as detector construction, physics lists, and primary particle generation—work together.

The main goal of the B1 example is to give learners hands-on practice with compiling and running a simulation, observing particle interactions, and producing output that can be visualized. It acts as a gentle introduction before progressing to more advanced Geant4 examples with detailed detectors and complex physics processes.

2.3 | Running the Example

After successfully compiling the B1 example, it can be executed from within the build directory. Running the produced executable opens the Geant4 interactive session, where the user can input commands to control the simulation. The basic procedure starts with the command `/run/initialize`, which prepares the geometry and physics processes, followed by `/run/beamOn N`, where `N` specifies the number of primary particles to simulate. This simple workflow allows users to observe how events are generated and tracked within the detector.

For visualization, Geant4 provides the following useful commands:

```
/vis/open OGL           % Start an OpenGL visualization window
/vis/drawVolume          % Display the detector geometry
/vis/scene/add/trajectories % Show particle tracks
/vis/scene/endOfEventAction % Draw tracks after each event
```

These steps give immediate feedback on how particles interact with the detector and form the basis for learning more advanced visualization and output features.

2.4 | Output of B1 Simulation

The output of the simulation consists of two main components. The first is the visualization output, which provides a graphical representation of the particle trajectories within the defined geometry. This allows us to visually track the paths of primary and secondary particles as they move through different volumes, showing how they interact with the materials, scatter, and deposit energy. The second component is the numerical data output, which was obtained by modifying the simulation code to record detailed information at each step of the particle tracks. This includes parameters such as particle type, process type, energy deposition, kinetic energy, spatial coordinates, momentum components, and track identifiers. Together, these two types of outputs provide a comprehensive understanding of both the visual behavior of particles in the simulation and the quantitative data required for further analysis and interpretation.

The figure 2.4.1 illustrates the particle trajectories in the Geant4 B1 example, where primary gamma particles and their secondary products are tracked as they traverse two distinct volumes. Shape1 has a trapezoidal geometry, while Shape2 is conical, representing different material regions or detector components. The visualization shows how particles interact with the materials, undergoing processes such as scattering, ionization, and conversion, producing secondary electrons, positrons, and gammas. Each particle's path is individually tracked, revealing detailed spatial information about their motion, energy deposition, and changes in direction. The Geant4 visualization window provides a comprehensive view of the geometry and particle tracks, allowing precise analysis of particle behavior in both shapes and the effects of the defined materials on particle transport.

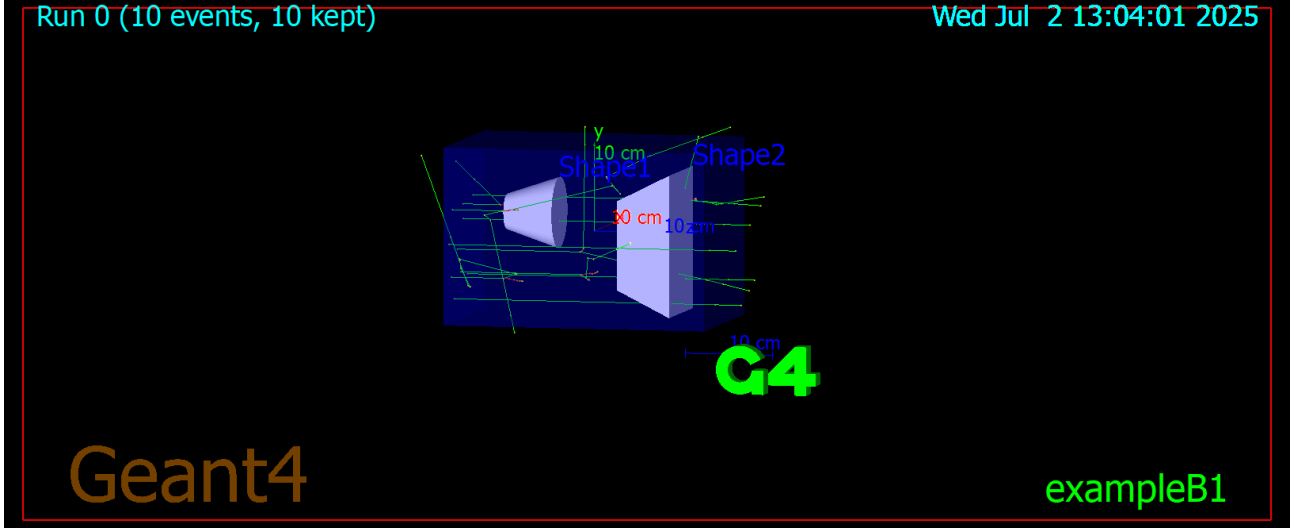


Figure 2.4.1: Particle trajectories in two different Geant4 volumes.

The table 2.4.1 below presents the simulation data obtained by shooting 10 gamma particles, each with an initial energy of 6 MeV, through a defined volume. Each row corresponds to a particle at a specific step in its trajectory, recording both primary and secondary particles. The data includes the type of particle, the process occurring at that step (such as conversion, Compton scattering, ionization, Bremsstrahlung, annihilation, or photoproduction), the energy deposited in the medium, the remaining kinetic energy, and the spatial coordinates in the X, Y, and Z directions. In addition, the table provides the momentum components, step derivatives of position, and unique identifiers for each particle track along with the corresponding parent particle ID. The recorded volume indicates in which medium the particle is present, while the time column shows the moment at which the particle reaches that step. This dataset allows detailed analysis of the particle interactions, the development of secondary particle showers, energy deposition patterns, and the evolution of particle trajectories. By examining the relationships between parent and secondary particles, the momentum changes, and the spatial-temporal distribution, one can gain insight into the transport and ionization effects of gamma radiation in the simulated environment.

Particle	Process	edep [keV]	KineticE [keV]	x [mm]	y [mm]	z [mm]	px [MeV]	py [MeV]	pz [MeV]	dx	dy	dz	TrackID	ParentID	Volume	Time [ns]
gamma	conv	0	6000	24.117	-0.803	40	0	0	6	0	0	1	1	0	Shape2	0.634
e+	eBrem	377.325	3743.421	24.117	-0.803	48.395	-0.171	-0.297	4.21	-0.041	-0.07	0.997	3	1	Shape2	0.662
e+	eloni	627.619	1050.354	24.261	-1.107	49.849	0.334	-0.355	1.392	0.227	-0.241	0.944	3	1	Shape2	0.667
e+	eloni	422.735	422.735	24.551	-1.415	51.056	0.193	0.086	0.752	0.247	0.11	0.963	3	1	Shape2	0.671
e+	annihil	0	0	24.648	-1.372	51.435	0	0	0	0.247	0.11	0.963	3	1	Shape2	0.673
gamma	compt	0	510.999	24.648	-1.372	51.435	0.079	-0.216	0.456	0.154	-0.423	0.893	6	3	Shape2	0.673
gamma	Transportation	0	239.833	26.214	-5.669	60.503	-0.184	-0.138	-0.069	-0.765	-0.576	-0.287	6	3	Shape2	0.707
e-	eloni	271.166	271.166	26.214	-5.669	60.503	0.262	-0.078	0.525	0.443	-0.132	0.887	7	6	Shape2	0.707
gamma	Transportation	0	510.999	24.648	-1.372	51.435	-0.079	0.216	-0.456	-0.154	0.423	-0.893	5	3	Shape2	0.673
gamma	compt	0	2315.742	24.261	-1.107	49.849	0.423	-0.695	2.168	0.182	-0.3	0.936	4	3	Shape2	0.667
gamma	Transportation	0	2192.467	26.102	-4.135	59.297	0.372	-0.978	1.927	0.169	-0.446	0.879	4	3	Shape2	0.7
e-	eloni	123.275	123.275	26.102	-4.135	59.297	0.051	0.283	0.242	0.136	0.754	0.643	19	4	Shape2	0.7
e-	eBrem	59.622	1234.581	24.117	-0.803	48.395	0.443	0.769	1.413	0.265	0.461	0.847	2	1	Shape2	0.662
e-	eloni	682.184	1167.756	24.174	-0.704	48.576	0.359	0.458	1.49	0.225	0.286	0.931	2	1	Shape2	0.663
e-	eloni	485.571	485.571	24.484	-0.309	49.862	0.636	0.5	0.28	0.743	0.584	0.327	2	1	Shape2	0.667
gamma	phot	4.041	7.204	24.174	-0.704	48.576	0	0.001	0.007	0.046	0.173	0.984	20	2	Shape2	0.663
e-	eloni	3.163	3.163	24.174	-0.703	48.58	-0.043	-0.037	-0.001	-0.755	-0.655	-0.025	21	20	Shape2	0.663

Table 2.4.1: Output obtained by shooting 10 gamma particles of certain energy.

Part II

CORSIKA

Chapter 3

Setup and Methodology

3.1 | Introduction to CORSIKA

Cosmic rays are high-energy particles, mainly protons and heavier nuclei, that enter the Earth's atmosphere from outer space. When they collide with atmospheric nuclei, they produce cascades of secondary particles called extensive air showers. Studying these showers helps in understanding high-energy astrophysics, particle physics, and atmospheric processes.

CORSIKA (COsmic Ray SIMulations for KAscade) is a Monte Carlo simulation program developed to model the evolution of these air showers. It tracks the interactions and propagation of particles through the atmosphere, including processes such as hadronic interactions, electromagnetic cascades, and ionization losses.

It allows researchers to simulate showers over a wide range of primary energies, particle types, and atmospheric conditions, providing detailed insight into the development and characteristics of secondary particles.

3.2 | Installation and Access

To use CORSIKA, it is necessary to first obtain access from the developers, as the program is not publicly available for direct download. A formal request must be made, after which permission and download links are provided.

CORSIKA runs on Linux systems, so familiarity with Linux commands and environment is essential. For this project, basic Linux skills were learned, including navigating directories, compiling programs, setting environment variables, and running scripts from the terminal. These skills are crucial for successfully installing and executing CORSIKA simulations.

Once access is granted, the software package is downloaded and extracted. The installation involves compiling the source code with a suitable Fortran compiler and ensuring all dependencies, such as necessary libraries for particle interaction models, are available. After compilation, test runs are performed to verify that the installation is successful and that the program runs without errors.

3.3 | Input File and Running CORSIKA

The input file in CORSIKA defines the parameters for simulating cosmic ray showers. It contains all the necessary information to configure a run, including the type of primary particle, its energy, and the direction of incidence. In this project, the input file was set up to run 10 showers for each configuration to allow clear visualization of all particle paths.

Key parameters in the input file include:

- **Primary particle type:** proton, muon, or other nuclei.
- **Primary energy:** the energy of incoming cosmic rays, set to match realistic cosmic ray spectra.
- **Zenith and azimuth angles:** determine the direction of the incoming particle relative to the atmosphere.
- **Atmosphere model:** defines the density and layers of the atmosphere for accurate particle propagation.
- **Interaction models:** specify hadronic and electromagnetic interaction models to simulate particle collisions.
- **Number of showers:** 10 showers were chosen for this project to track multiple particles simultaneously.
- **Observation levels:** altitudes at which particle data is recorded.
- **Random seed:** ensures reproducibility of results.

Running CORSIKA involves executing the program from the terminal with the input file as an argument. The simulation reads these parameters, propagates the particles through the atmosphere, and generates output files containing detailed information about all secondary particles, energy deposition, and shower characteristics.

This setup allows both a broad view of shower development and detailed analysis of individual particle paths, which is essential for studying ionization in different media.

3.4 | How CORSIKA Simulation Works

CORSIKA simulates the development of extensive air showers using a Monte Carlo approach. The simulation begins with a primary cosmic ray entering the atmosphere, and then follows its interactions with atmospheric nuclei. Each interaction generates secondary particles, which are propagated further, creating a cascade of particles through successive generations.

Key aspects of the simulation include:

- **Interaction models:** CORSIKA uses specific hadronic and electromagnetic models to calculate particle collisions and decays. These models determine how energy is transferred and how secondary particles are produced.

- **Atmosphere layers:** The program divides the atmosphere into layers with varying density and composition to accurately simulate particle propagation and energy loss.
- **Shower development:** The program tracks the position, energy, and type of each particle at multiple observation levels, allowing reconstruction of the longitudinal and lateral structure of the shower.
- **Particle tracking:** Individual particles can be followed to study their paths, interactions, and energy deposition in different materials.
- **Thinning and statistical weights:** To manage computational resources, low-energy particles can be thinned, with corresponding statistical weights applied to maintain accurate overall energy distributions.

This workflow provides a detailed picture of how primary cosmic rays initiate showers and how energy is distributed among secondary particles in the atmosphere. It forms the foundation for analyzing shower characteristics, energy deposition, and ionization effects in different media.

Chapter 4

Results and Discussion

4.1 | Obtaining and Processing Output Files

The initial output files from CORSIKA are in a proprietary binary format, which is an inbuilt format designed for efficient storage of large simulation data, including detailed information about particle types, energies, positions, and interactions. This format cannot be opened or interpreted by standard spreadsheet applications and requires specialized processing to extract meaningful data.

At first, only basic text editors could be used to view the raw binary files, which appeared as unreadable characters. Multiple attempts were made to convert the binary files using common conversion tools and methods, but none provided usable output. This highlighted the need for a systematic approach to decode and structure the data.

Python was then employed as a solution. Using the Jupyter Notebook environment, several scripts were developed to parse the binary data. Key Python libraries used included:

- **NumPy:** for handling numerical arrays and processing large datasets.
- **Pandas:** for structuring the data into tabular form and facilitating conversion to CSV.
- **Struct:** for interpreting the binary file according to the CORSIKA data specification.

Through iterative testing and refinement, the binary files were successfully converted into a readable, structured format suitable for detailed analysis. This process provided an organized dataset, forming the basis for subsequent.

Here is a sample of the output data obtained from the CORSIKA binary file after conversion to a readable format using Python. Each row represents a particle detected in the simulation with its momentum, position, and arrival time.

id	p_x	p_y	p_z	x	y	t
1011	-0.01404	-0.02179	0.09257	-3713.18	21414.51	370409.06
1011	-0.00236	-0.00909	0.02615	4553.10	-3084.65	370578.34
1011	-0.00301	-0.00752	0.02443	4408.37	-2053.30	370567.03
1011	-0.00348	-0.00527	0.01519	-2267.87	-4657.10	370630.44
1011	-0.00049	-0.00734	0.02386	11031.85	-3027.52	370570.38
1011	-0.00041	-0.00398	0.01537	-5406.10	219.14	370595.72
1011	-0.00145	-0.00370	0.01244	-963.44	-805.46	370580.30
1011	-0.00189	-0.00575	0.01851	-41.39	-1664.94	370585.44
1011	-0.01337	-0.05831	0.16534	1038.42	-4197.65	370608.84
1011	-0.00613	-0.00989	0.04129	-2255.96	17028.75	370433.28
1011	-0.00600	-0.00706	0.02709	-19991.98	6118.39	370636.25
1011	-0.00791	-0.00434	0.02921	-39781.73	52068.10	370464.60
1011	-0.00492	-0.00325	0.01149	-29235.50	12644.74	370676.16
1011	-0.00598	-0.00762	0.02996	-17015.35	13047.21	370543.38
1011	-0.01076	-0.00599	0.04608	-21477.08	28162.80	370478.00
3011	-0.00408	-0.00564	0.01157	-23978.18	29296.99	370516.56

Sample output data from CORSIKA showing particle ID, momentum components (p_x , p_y , p_z), position coordinates (x , y), and time t .

4.2 | Working with Long Files and CSV Conversion

After successfully converting the binary output files, focus shifted to other CORSIKA outputs, particularly the "long file". The long file contains comprehensive data on the secondary particles generated in each shower, including their types, energies, positions, arrival times, and trajectories at multiple observation levels. This file is essential for detailed studies of shower development and particle propagation.

The long file is generated directly by CORSIKA when specified in the input parameters. Obtaining it involves enabling the appropriate output option in the input card and running the simulation, which produces a large dataset that cannot be easily read in its raw form due to its inbuilt CORSIKA format.

To make the long file usable, Python scripts were again employed. Using the NumPy, Pandas, and Struct libraries, the data was parsed and converted into a structured format. The process included:

- Reading the binary content and decoding the records according to the long file specification.
- Organizing particle data into arrays and tables with clear headings for each parameter.
- Converting the structured data into CSV files for easy inspection, filtering, and further analysis.

This conversion enables researchers to analyze shower characteristics efficiently, create plots of particle distributions, and perform statistical studies. By having the long file in CSV format, it becomes straightforward to integrate the data into Jupyter Notebooks or other analytical tools, ensuring a comprehensive examination of the simulation results.

Here we present sample long-file output data obtained from the CORSIKA simulation. The first table shows particle counts at different observation depths for various particle types, while the second table shows energy deposit contributions from different processes at corresponding depths.

DEPTH	γ	e^+	e^-	μ^+	μ^-	H	Ch	N	Chk
10	0	0	0	0	0	1	1	0	0
40	59	8	9	11	12	62	96	0	0
70	253	50	61	35	42	42	224	0	0
100	1103	175	221	69	78	105	629	0	0
130	3498	570	716	155	156	169	1717	0	0
160	8534	1312	1746	246	227	223	3688	0	0
190	17452	2512	3378	296	303	319	6720	0	0

Sample output data from CORSIKA showing particle counts at different observation depths for a single shower. Columns represent depth (g/cm²), gammas, positrons, electrons, muons, hadrons, charged particles, nuclei, and Cherenkov photons.

DEPTH	γ	EM-I	EM-C	μ -I	μ -C	H-I	H-C	ν	Σ
5	0	0	0	0	0	0.04706	0	0	0.04706
35	0.00111	0.21285	0.03165	0.33393	0	1.12274	0.43633	239.40	241.54
75	0.04682	3.66332	0.34212	2.12105	0	1.54176	1.97132	69.38	79.06
115	0.50397	20.6183	1.8371	5.9925	0	3.02512	3.60481	162.70	198.29
145	1.20931	55.3705	5.4216	10.2117	0	4.14177	10.113	108.29	194.76
165	2.20731	88.8812	8.9364	12.5703	0.11066	4.93216	7.2051	102.22	227.07
175	2.31303	111.343	11.5948	13.6344	0	5.77283	7.4220	148.83	300.91

Sample output data from CORSIKA showing energy deposition from different processes at various depths for a single shower. Columns show contributions from electromagnetic (EM), muonic (μ), and hadronic (H) components. The suffix "I" denotes ionization energy, while "C" denotes energy lost due to simulation cuts. Also includes neutrino (ν) contributions and the total sum (Σ).

Chapter 5

Experience and Skills Acquired

5.1 | Overall Internship Summary

During this internship, a comprehensive understanding of particle simulation and analysis tools was achieved. The work included studying and gaining practical experience with both Geant4 and CORSIKA simulation toolkits.

Key learning outcomes include:

- **Geant4 Basics:** Understanding the structure and workflow of Geant4 simulations, including defining geometry, materials, particle sources, and tracking particle interactions. Learned to visualize particle tracks and extract energy deposition data.
- **CORSIKA Fundamentals:** Gained hands-on experience in simulating cosmic ray air showers, setting up input files, configuring primary particles and energies, and interpreting outputs such as binary files and long files.
- **Data Processing Skills:** Developed proficiency in Python programming for data conversion, using libraries such as NumPy, Pandas, and Struct to process and analyze large datasets from both Geant4 and CORSIKA.
- **Linux Proficiency:** Learned essential Linux commands and environment management for installing, compiling, and running simulation software.
- **Analysis and Visualization:** Practiced converting complex outputs into readable formats, creating CSV files, visualizing particle distributions, energy deposition, and shower profiles.

Overall, the internship provided practical experience in high-energy particle simulations, enhanced programming and data analysis skills, and a solid foundation for future research in cosmic ray physics and computational modeling.

Chapter 6

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